

MID-SEMESTER EXAMINATION, SESSION: 2023-24

Examination & Semester: 8th and 6th B.Tech (Engineering)

Subject: Introduction to Geographical Information System (MIND409)

Prof Girind Vilas

Note: Answer all the questions.

Time: 2 hr

Q.No.	Question	Marks
1.	Define Remote sensing. Explain briefly about the process of remote sensing cycle with a neat supporting diagram.	6
2.	What are the advantages and limitations of Geomatics engineering? What is geometric rectification of satellite images?	6
3.	With a neat diagram explain the spectral reflectance curves for vegetation, soil and water.	6
4.	Define GIS. List the typical GIS data sets with classification of each.	6
5.	a) What is meant by active remote sensing and passive remote sensing? b) Define atmospheric window? What are the atmospheric window wavelength? c) List the various applications of Geomatics in engineering and science field. d) Elaborate GNSS survey.	4x1=4

Prof Govind Villuri

END-SEMESTER EXAMINATION, SESSION: 2023-24
Examination & Semester: 6th and 8th B.Tech

Subject: Introduction to Geographical Information System (MND409)

Note: Answer any five questions.

Time: 3 hr

Q.No.	Question	Marks
1.	a) Define Georeferencing? What are the advantage of Georeferencing? b) List and explain various data inputs to GIS. c) Explain how spatial data and attribute data integrated to make a GIS. d) Define GIS. Explicit the key components of GIS.	4x5=20
2.	a) What is spatial analysis in GIS? Brief the query based analysis with examples. b) Elaborate the various database structures in GIS.	2x10=20
3.	a) List the map projections and how Universal Transverse Mercator (UTM) Works. b) What is mosaic datasets? Brief manual mosaic of satellite imagery for GIS map preparation.	2x10=20
4.	What is multiple layer analysis in GIS and explain the multiple layer analysis concept with any two example.	20
5.	a) List and explain elements of Raster Data Model and Vector Data Model. b) Explain the topological data structure of vector data with neat sketch.	2x10=20
6.	Write short note on : a) Georeferencing b) Editing in GIS c) Topological errors in conversion. d) Geographical coordinate system	4x5=20

$\frac{13}{36}$

Distance

Prof Bhaskara Behara

Indian Institute of Technology (Indian School of Mines) Dhanbad

Department of Mining Engineering

Mid-Semester Examination (Winter Session) 2023-24 for B.Tech. Students (45 nos.)

Subject: Advanced Underground Mining Methods (MND 411) Venue: NLHC LH 08

Time: 120 Minutes (08.00 AM-10.00 AM) Date: 02.03.2024 Maximum Marks: 64

Answer questions of all the sections

Question No.	Section A	Marks
	Answer any four questions [4×10=40 Marks]	
1.	Discuss the operational challenges associated with thick seam mining. Enumerate the typical ground control requirements for mining thick seams.	[10]
2.	What are the different factors that govern the choice of face length in Longwall workings? Why it is critical to operate with optimal face length for a given mining condition?	[10]
3.	a. What are the different zones that form above the coal seam with the progressive mining in a typical longwall working? How do the dynamics of caving height alter with the variation of bulking factor and mining height? b. Discuss the supporting cycle and loading characteristics of the powered roof support deployed in the longwall operation.	[5+5]
4.	With neat sketches, describe the abutment stress distribution about the longwall panel.	[10]
5.	a. What are the typical loading scenarios in the longwall workings? b. Discuss the mechanism of face damage and various forms of damage parameters.	[3+7]

Section B

Answer all the questions [2×12=24 Marks]

7. a. Estimate the different roof layers above the coal seams for a longwall panel of Mine 'A' by applying the Roof Separation Index (RSI) approach (Refer to Table 1), where there is no influence of groundwater on the working. (Refer to Roof Separation Index (RSI) Table, Table 2, provided in the question paper for assignment of different ratings associated with RSI parameters) [5+7]
- b. Applying the concepts of Beam and Cantilever models, estimate the spans of the first local fall, subsequent local fall, main fall and periodic caving pertained to the working of Mine 'A'.

3015
120
82

8.

a. A longwall face of Mine 'B' is planned to operate under the cover depth of 150m with a face length and face height of 180m and 3.5m, respectively. The Physico-Mechanical properties of overlying roof rocks related to the working are given in Table 3. Estimate the capacity of powered roof support that needs to be deployed in the given condition considering the canopy length of 4m, canopy width of 1.75m and web depth of 0.8m for the working. Below are some useful relations for solving this question:

[7+5]

$$C_m = \left(\frac{1}{P_b} + 9.6h_s \right) + \left(\frac{K_1}{K_1 + 1.5} - 23 \right)$$

$$\text{Support Capacity (T)} = P_r \times W_c \times (L_c + W_d + 0.3)$$

b. Keeping the other parameters constant, If the mine management planned to operate the longwall face with a 4.5m face, then what would be the support capacity for altering the face height from 3.5m to 4.5m?

Table 2: Roof Separation Index (R.S.I) Table

Sl. No.	Parameter	Unit/ Rating	Rating distribution				
			<5	5-25	25-50	>50	
1	Uni-axial compressive strength	MPa Rating	<5 1	5-25 2	25-50 4	>50 7	
2	RQD	% Rating	<25 3	25-50 8	50-75 13	75-90 17	>90 20
3	Average core length	cm Rating	<5 5	5-30 10	30-100 20	100-300 25	
4	Ground water condition/wetness of strata	Condition Rating	Dry 10		Watery formation 0		
5	Bed thickness	m Rating	<1.5 -28	1.5-2.5 -16	2.5-3.5 -12	3.5-4.5 -5	>=4.5 0
6	Rock formation of bed	Rock type Rating	Sandstone 0		Others (shale, coal, clay, shaly coal) -25		

Prof Bhaskara Behara

Indian Institute of Technology (Indian School of Mines) Dhanbad

Department of Mining Engineering

End-Semester Examination (Winter Session) 2023-24 for B.Tech. Students (45 nos.)

Subject: Advanced Underground Mining Methods (MND 411) Venue: NLHC LH 07

Time: 180 Minutes (04.00-07.00 PM) Date: 05.05.2024 Maximum Marks: 96

Answer questions of all the sections

Question No.	Section A Answer any four questions [14×4=56 Marks]	Marks
1. UGC pg13 UGC PG16	a. Describe the CRIP method in the UCG application with the schematic diagram. b. Discuss the various factors affecting the process performance in UCG technology.	[4+10]
2. Highwall Mining pg 7 ;16; 17 chart gpt	a. Discuss the applicability, working methods and mechanism vis-à-vis major infrastructural needs for successful deployment of Highwall Mining Technology. b. Briefly enumerate the technological advancement witnessed in longwall technology at the Adriyala Longwall Project (ALP) of SCCL Mine.	[8+6]
3.	What are the different slicing methods available to mine thick coal seams? Compare the inclined slicing method in ascending to descending order concerning their applicability, mode of development and depillaring in a given mining condition.	[3+11]
4. pg-176 metal pdf pg 166 metal pdf	a. What is Livingston's crater theory? Why it is so important to the VCR method? b. Discuss the various failure modes of the pillar with correct reasoning and schematics.	[7+7]
5. pg 40 metal pdf	a. Provide a comparative note on Empirical Vs. Qualitative method for selection of a mining method. b. Select the appropriate mining method from the ore deposit data given in Table 2 using the Nicholas Approach. (Refer Table 3-6 for the application of Nicholas Ranking Method)	[6+8]
Section B Answer all the questions [2×12=24 Marks]		
6.	a. A longwall face of Mine 'X' is planned to operate under 350m cover depth with a face length of 200m and a face height of 3.5m. The physico-mechanical properties of the various roof layers are given in Table 1. Estimate the span of the main fall for the given mining condition using the relation:	[15+5+4]

Baskara
Wright
Ardhan

2023 UGC

$$E = (0.31 \times 5c) GPa$$

2023 3GPa

Table headings

Figure - Overlay of GCPs on High resolution satellite image data (Copernicus website)

Bullet

Figure showing slow chat processing ins NAA

$$W_s = 324 F_L^{-0.43} D^{-1.38} \sqrt{\frac{\sigma_{mIR} \times T_{IR}}{\gamma_{IR}}} \sqrt{\frac{\sigma_{mMR} \times T_{MR}}{\gamma_{MR}}}$$

Figure: Map showing of Sentinel data overlay on

- Following the same strata characteristics if the mine management plans to operate the next three longwall faces by changing the face length to 225m, 275m, and 300m, then what would be the span of the main fall for these new panels?
- In the given relation, why are we not considering the variable pertaining to coal strength and mining height for the determination of the main fall span in the given geo-mining condition?

Table heading

7.

- A room and pillar operation is planned for a depth of 450m in a stratum that has a compressive strength estimated as 26.37 MPa. Entries and crosscuts are planned at 6.5m. Determine the possible extraction ratio. (Assume, $\gamma = 0.023$ MPa/m)
- If the pillars are square in size, for the estimated extraction ratio in (a), determine the pillar dimension.
- Suppose the pillar safety factor of 1.5 is required for the room and pillar method with the square pillar, determine the extraction ratio and pillar dimension.
- With the same condition (refer to conditions, (a) and (c)), suppose the pillars are three times as long as they are wide, determine the extraction ratio and pillar dimension.

[3+3+5+5]

Processing of SAR Data

Bullet pt Figure - Gradient Displacement

Study Area Points
Ground Control pts
Subsidence Monitoring
Source

Ground, 2022

Table 1: Physico-mechanical characteristics of the roof layers

Layers	Thickness (m)	Density (Kg/m ³)	RQD (%)	Compressive Strength (MPa)	Tensile Strength (MPa)
Coal	3.5	1400	40	23.16	1.54
Immediate Roof	35.23	2045	80	12.25	1.10
Main Roof	17.46	2050	76	12.89	1.13
Overburden	297.31	2089	74	13.21	1.26

Assume, $G = 0.03^\circ\text{C/m}$, $\beta = 8 \times 10^{-6}/^\circ\text{m}$, $\nu = 0.25$, $\gamma = 0.023$ MPa/m

Map showing the Study Area the JCF

Slide: 2 -> Bullet points

3 -> Figure 1

Slider Bullet points

Objectives -> 1 page

1 -> Objective 1
Topography
Elevation
Working Methodology of Radar S

Topography of Study Area

Map of Jharia Region using Sentinel data